1. Identity and decision problem

Musanze Harvest Intelligence Service

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Assessment / signature	Individual Assignment 2 / AI-A2-7343
Deadline	19 October 2026, 5:00 PM CAT

A Musanze potato cooperative needs to consolidate inconsistent farm records into estimates that support harvest collection and field visits. The field officer remains the decision maker.

Task	Question and output
Regression	How many tons will this farm harvest? Output: predicted_yield_tons.
Classification	Is synthetic yield per hectare below the base-data median? Output: high-risk label and probability.
Clustering	Which farms have similar measured conditions? Output: cluster membership.

Ethical boundary

This is a classroom prototype trained entirely on synthetic data. It must not replace an agronomist or decide access to finance, insurance or public services. A high-risk label is a dataset definition, not a diagnosis of crop disease or a verified forecast of farm failure.

AI-assisted implementation and draft explanations are disclosed in evidence/AI_USE.md. The interpretations and reflection in this report are taken from docs/PERSONAL_REVIEW.md, which the student confirmed as completed and reviewed. No signature or authentic demonstration is certified by this report update.

2. Data and reproducibility

Column	Meaning / range
rainfall_mm	600 to 1400; numeric
avg_temp_c	12 to 26; numeric
soil_moisture_pct	20 to 80; numeric
fertilizer_kg	0 to 250; numeric
field_size_ha	0.2 to 4; numeric
cooperative_support	0 to 1; binary
season	A or B
yield_tons	Synthetic harvest target; minimum 0.50 tons
high_risk	1 if pre-missingness yield/ha is below base median; otherwise 0

Seed = last four registration digits = 7343. numpy.random.default_rng(7343); base rows = 240 + (7343 % 41) = 244. Labels are created before missingness, using unrounded generated values. The median is part of the specified synthetic label generator, not a fitted prediction-time feature.

Yield = 0.004 rainfall + 0.06 moisture + 0.02 fertilizer + 1.30 field size - 0.25 abs(temperature - 18) + 0.70 support + season effect + N(0, 0.80). Season A adds 0.50; B subtracts 0.20. Clip at 0.50.

Raw rows 247; duplicates 3; cleaned rows 244. Exactly 12 unique positions per missingness column are sampled separately (overlap is allowed). The first three corrupted records are appended as exact duplicates. Clean missing counts: moisture 12, fertilizer 12; raw counts: moisture 12, fertilizer 12.

CSV floats have four decimals; UTF-8 with explicit LF line endings. SHA-256 hashes final CSV bytes:

19a1558643adbce326107d7fcfb9df4a
8cd48cf745f2af43b8e07af6b6975d62

Two complete pipeline executions produced identical CSV fingerprint, metric JSON files, cluster report and test-prediction CSV bytes. See evidence/reproducibility.json. Missing predictors are preserved until training-only median imputation.

3. Pipeline design and leakage controls

After removing duplicate records, a fixed stratified 75/25 split produces 183 training and 61 testing farms; random_state=7343. Original zero-based CSV positions remain row IDs. Duplicate removal precedes the split so copies cannot appear on opposite sides.

Stage	Transformation / control
Raw CSV -> validation	Exact schema, numeric bounds, categorical values, missingness and duplicate evidence.
Clean -> split	No imputation, scaling or encoding is fitted before splitting.
Numeric branch	Training medians -> StandardScaler fitted on training inputs. Binary support is scaled with other numeric inputs.
Season branch	OneHotEncoder fitted on training season values; inference accepts A/B only.
Model -> saved pipeline	Each candidate has an independent ColumnTransformer. Final joblib pipelines retain their fitted transformers.

Only rainfall, temperature, moisture, fertilizer, field size, cooperative support and season are inputs. An explicit whitelist excludes yield_tons, high_risk and any yield-per-hectare column. Clustering uses the same measured-input whitelist and training farms only.

Regression candidates: mean DummyRegressor, LinearRegression, and RandomForestRegressor (200 trees). Classification: most-frequent DummyClassifier, LogisticRegression and RandomForestClassifier (200 trees). Forests use one worker; random states are fixed. KMeans tests K=2..6 with n_init=20.

Selection uses lowest test RMSE for regression and highest test F1 for classification (recall breaks ties). The PDF permits selection on test evidence, but this makes the selected score optimistic as an estimate of future performance. Future work should use train/validation selection and a separate untouched test set.

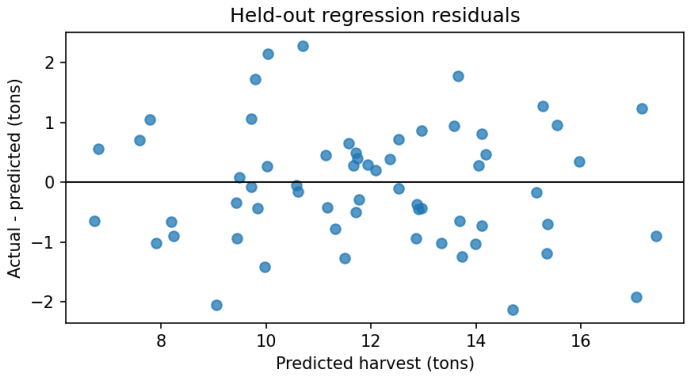
Training -> artifacts/models -> shared Predictor -> CLI and FastAPI. Inference never retrains; the API loads saved trusted pipelines at startup. Tests verify stored imputer medians against training medians and confirm feature exclusion.

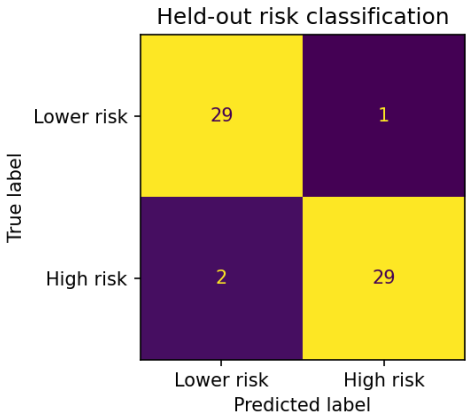
4. Regression and classification

Regressor	MAE	RMSE	R squared
DummyRegressor	2.1753	2.7148	-0.0147
LinearRegression	0.7967	0.9709	0.8702
RandomForestRegressor	0.9501	1.2217	0.7945

Classifier	Precision	Recall	F1	ROC-AUC
DummyClassifier	0.0000	0.0000	0.0000	0.5000
LogisticRegression	0.9667	0.9355	0.9508	0.9892
RandomForestClassifier	0.8966	0.8387	0.8667	0.9769

Selected: LinearRegression (lowest RMSE); LogisticRegression (highest F1). Positive class = 1, probability threshold = 0.5. All candidates share the same held-out farms.





Residual sign in the plot is actual minus predicted; the CSV uses predicted minus actual. Confusion-matrix rows are actual classes, columns predicted classes. ROC-AUC uses probabilities and is reported only with both true classes.

In row 2, the actual harvest was 12.9863 tons, while the model predicted 10.7003 tons. This gives an error of -2.2860 tons. The negative value means that the model predicted a smaller harvest than the actual harvest.

This error is important because farmers and cooperatives in Musanze may use harvest predictions to plan transport, storage and sales. If the prediction is too low, they might not prepare enough transport or storage space.

To understand this error better, I would check the rainfall, temperature and field size measurements. These values can affect harvest predictions. I would also compare the prediction with actual farm records before using it for important decisions.

In row 28, the actual risk class was 1, but the model predicted 0, with a high-risk probability of 0.4500. This is called a false negative because the model failed to identify a farm that was actually high risk.

This mistake could be serious because an agricultural officer might think the farm is safe and fail to provide support on time.

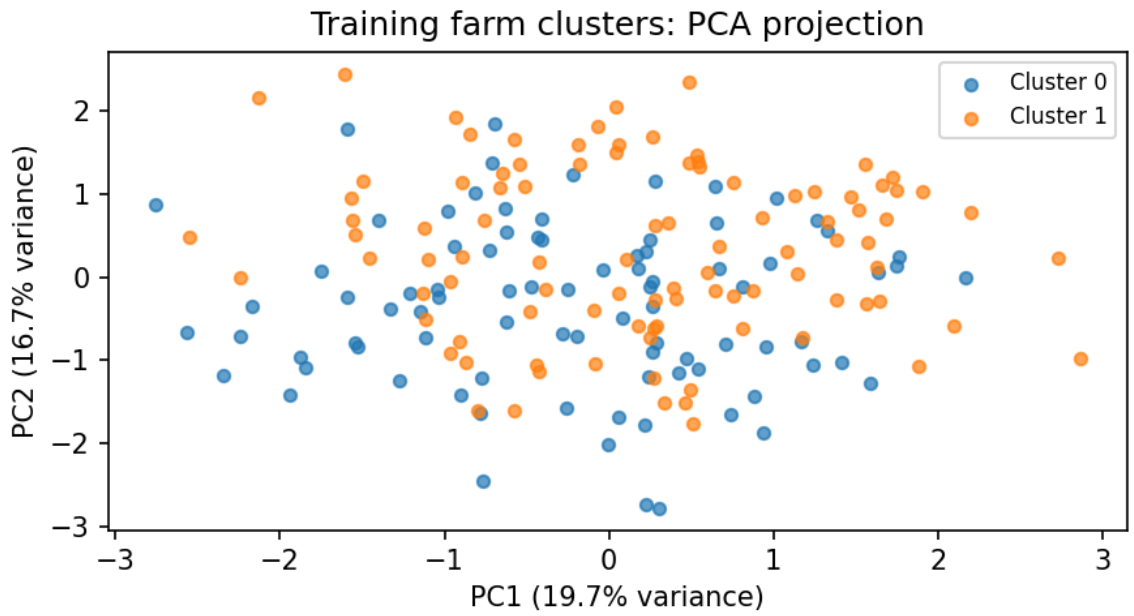
For this reason, I think AI predictions should support human decisions instead of replacing them. Agricultural officers should review uncertain predictions and check the farm's actual condition before making a final decision.

High risk here means below-median synthetic yield per hectare, not a verified crop-failure event.

5. Clustering and operational meaning

K	Training silhouette
2	0.15659
3	0.13650
4	0.13602
5	0.13809
6	0.13716

Selected K=2 maximizes silhouette among the five candidates. The low best score (0.157) indicates weak separation. Here the groups split exactly by cooperative support; other mean conditions are similar. Silhouette does not establish agronomic validity. Cluster IDs are arbitrary; no target enters fitting.



The PCA projection is fitted on training inputs; axes show the explained variance of each principal component. Separation in two dimensions may conceal overlap in the full feature space.

Training group 0: 86 farms; mean rainfall 1017.9 mm, temperature 19.7 C, field size 2.01 ha and support fraction 1.00. Use these measured similarities for field-visit planning, subject to officer review.

Training group 1: 97 farms; mean rainfall 990.0 mm, temperature 19.5 C, field size 2.01 ha and support fraction 0.00. Use these measured similarities for field-visit planning, subject to officer review.

Profiles use observed numeric means (missing measurements omitted) and include season proportions in the JSON. They do not identify causal interventions.

Cluster 0 contains 86 farms with similar characteristics. Their average rainfall was 1017.9 mm, average temperature was 19.7°C and average field size was 2.01 hectares. All farms in this training group had cooperative support.

This information could help agricultural cooperatives plan field visits and training activities. For example, officers could organize group training sessions for farmers with similar conditions instead of visiting each farm separately.

However, being in the same cluster does not mean that the farms have the same level of risk. Clustering only shows similarities between the farm records. Each farm still needs individual assessment when necessary.

6. API contract and actual testing

Endpoint	Contract
GET /health	200: status and models_ready. Missing artifacts produce degraded health.
GET /model-info	200: selected models, features, K, seed, verification code and fingerprint; 503 if unavailable.
POST /predict	Seven required bounded fields -> yield, risk label, probability, cluster and verification code. Invalid input -> 422 with field details; unavailable models -> 503.
GET /docs	Swagger UI; /openapi.json describes typed request and response.

Valid request: {"rainfall_mm":1000,"avg_temp_c":18,"soil_moisture_pct":50,"fertilizer_kg":120,"field_size_ha":2,"season":"A","cooperative_support":1}

Actual CLI response: {
"predicted_yield_tons": 12.4816924638537,
"predicted_high_risk": 0,
"high_risk_probability": 0.33931241711599996,
"cluster": 0,
"verification_code": "AI-A2-7343"
}

Invalid sample combines rainfall 1500, temperature "warm" and season C. CLI returned exit 2 with validation details. API tests exercise missing fields, invalid values and malformed JSON (HTTP 422). Numeric strings, boolean support, NaN and extra target fields are rejected by shared strict Pydantic validation.

Executed evidence

28 passed, 1 warning in 16.27s

pytest checks valid inference, missing/type/range/season/extra/nonfinite failures, reproducible bytes, 247/244 row counts, 12 missing values per cleaned column, signature, saved model loading, wrong seed rejection, leakage controls, train-only imputation, CLI behavior and FastAPI responses. Full raw stdout/stderr is in evidence/execution_log.json.

FastAPI TestClient verified health, model-info, docs, valid prediction, missing-field validation and malformed JSON. Explicit endpoint responses are saved in evidence/api_checks.json. No authentic video or student signature is claimed. No live browser interaction, load test or real agricultural validation is claimed here.

7. Responsible AI model card

Aspect	Scope and limitation
Intended user / use	Cooperative field officers; classroom harvest planning and prioritization support.
Data provenance	244 synthetic base farms. No actual farm measurements, farmer identities or consented field outcomes.
Target meaning	High risk means below-median synthetic tons/ha, not disease, poverty or an independently validated failure event.
Performance scope	One fixed split. Test evidence is also used for model selection. Small sample and random missingness constrain reliability.
Risks	Overconfident estimates, unfair service allocation, synthetic correlations, measurement errors and failure under distribution shift.
Prohibited use	Automatic finance, insurance or public-service access decisions; replacing an agronomist.
Human oversight	Confirm farm measurements, inspect uncertain cases and obtain agronomist advice before action. Avoid punitive interpretations of risk labels.
Security / privacy	No authentication or production rate limiting. Bind locally. Load only trusted joblib files, which can execute code. Collect minimal farmer data in any future deployment.

Improvement plan: gather representative real seasonal data with consent, verify measurement quality, use separate validation/test sets, test subgroup performance, quantify harvest uncertainty and calibrate risk probabilities. Compare richer temperature features under training-only cross-validation. Monitor drift and document retraining decisions.

Personal reflection

This project shows how artificial intelligence can support agricultural planning. Regression helps estimate harvest quantities, classification helps identify possible risks, and clustering helps group farms with similar characteristics.

One important lesson from the results is that AI models can make mistakes. For example, the regression model underestimated one harvest, and the classification model failed to identify one high-risk farm. This means AI predictions should be checked before they are used for important agricultural decisions.

One improvement I would recommend is adding a review process for farms with uncertain risk predictions. This would allow agricultural officers to examine those farms more carefully instead of relying only on the model.

I also understand that the project uses synthetic data. Therefore, the results are suitable for learning and demonstrating machine learning, but the system would need further testing with real agricultural data before practical use.

Source: supplied eight-page INES Ruhengeri SWE 3513 Individual Assignment 2 PDF, issued 16 October 2026. All numerical claims in this report are read from the accompanying generated artifacts.